PSTAT 10 Data Science Principles

Lecture 13: Databases

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Introduction

🔁 Review: Lectures 9 – 12

🤔 This week we will move away from the language `R` and instead focus on `SQL`.

- The following lectures will (explicitly) use minimal R.
- It is still important to be comfortable with topics such as simulation, random variables and Monte Carlo methods.

> 📌 The bulk of the final exam will focus on the language R. Assignment 5 and the last labs will also use R.

👀Outline: Lecture 13

👇In today's lecture we will introduce ourselves to larger, more complex database structures. We aim to answer the questions:

- What is a database?
- What is a relational database?
- What is a database management system (DBMS)?

Furthermore we will cover the relational model, both the:

- Structural portion; and
- Integrity portion

Databases

🗄️ What is a Database?

DEFINITION

A database is an organized collection of structured information, or data, typically stored electronically in a computer system.

- Medical records
- Bank accounts
- Credit histories
- Historical security prices
- Census data
- Airline schedules and bookings
- Bus and train timetables
- Student records
- Customer purchasing histories
- Medical trial data

> 📌 Essentially anything can be stored as a set of related data values.

🗄️Why Do We Use Databases?

We have to when datasets get incredibly large.

For example, a standard Excel document can store roughly:

- 1 million rows
- 16000 columns

Most Excel documents however are much smaller, around 0.5MB to 1MB.

> What happens if we are dealing with a very large dataset?

🗄️Example — Why Do We Use Databases?

Consider the `nycflights13` data from previous weeks.

```
1 library(nycflights13)
2 format(object.size(nycflights13::flights), units ="Mb")
```

```
[1] "38.8 Mb"
```

This is still a fairly small dataset. Really large datasets can be many (GB) or even (TB).

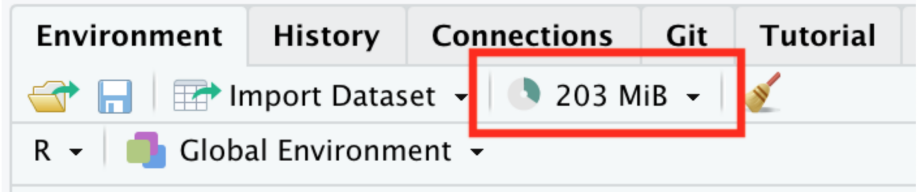
For example:

- High-frequency securities data
- Medical trials data
- Advertising data

💾Memory Usage

If we import a large dataset into R, then we are storing that data in our computer's memory.

In our environment tab in RStudio we can see a read-out of the amount of memory we are currently using.



Memory usage in RStudio

> 📌RStudio can only use so much memory before computer performance declines rapidly with high memory usage.

💪 Exercise — Available Memory

1. Determine how much random access memory (RAM) your computer has.
    - For Macs go to `Apple -> About This Mac -> Memory`
    - For Windows go to `Task Manager -> Performance -> Memory`
2. Determine how much memory `R` has available.
    - Explore `?gc`.
    - Run `gc()` (results will vary!)

03:00

✅ Solution — Available Memory

My Mac has 1TB of storage and 16GB of RAM (which is starting to lag behind 2026 laptops now).

```
1 gc()
```

```
         used (Mb) gc trigger  (Mb) limit (Mb) max used  (Mb)
Ncells  726192 38.8    1362828  72.8         NA  1362828  72.8
Vcells 6433266 49.1   21335211 162.8      16384 17398577 132.8
```

- The `used` columns report the memory `R` is currently occupying.
- The `max used` columns report the peak usage so far this session.
- Values are shown for both `Ncells` (objects) and `Vcells` (vectors/data).

Too Much Data

The table below details several databases that are too large to load into R.

Large Data File Examples

File Number	File Name	Size
1	Diabetic Retinopathy Detection	82.2 GiB
2	American Epilepsy Society Seizure Prediction Challenge	59.6 GiB
3	Avito Duplicate Ads Detection	48.4 GiB
4	Microsoft Malware Classification Challenge (BIG 2015)	35.3 GiB
5	GE Flight Question	34.4 GiB

🗄️Dealing with Big Data

Our approach:

- Store the data on an external disk
- Open a connection to the data
- Load portions of the data into local memory as needed
- The dataset stored on this disk is a database

> 📌We only ever hold what we need in memory — the connection lets us query a database far larger than our RAM.

📖Formal Definitions

We defined a database earlier as an **organized collection of structured information**, or data, typically stored electronically in a computer system.

Our databases will satisfy the following **two traits**:

1. They will be **machine-updatable** (i.e. a collection of variables).
2. They will be **logically coherent** (i.e. they will have some understandable organizational structure).

🖥️Database Management Systems (DBMS)

We manage databases through Database Management Systems (DBMS).

There are four main types:

- Hierarchical
- Network Model
- Relational Model
- Object-Oriented Model

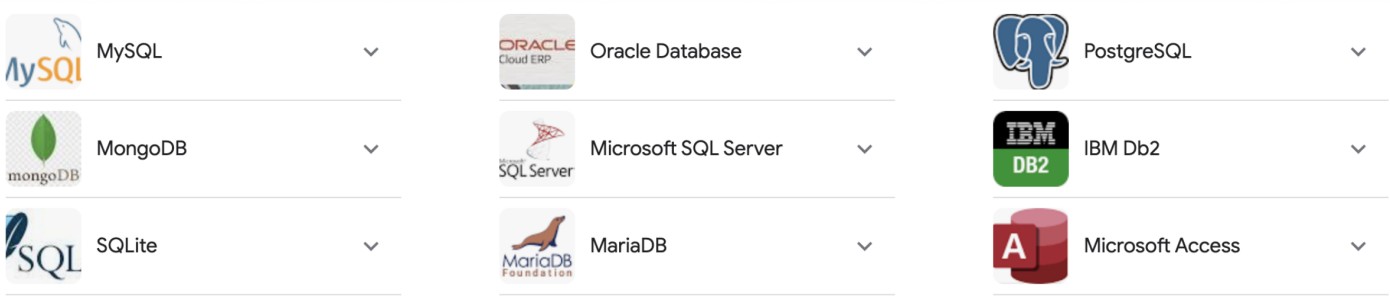
Broadly, we can classify DBMS into:

1. Relational Database Management Systems (RDBMS); and
2. Non-Relational Database Management Systems (NoSQL).

🖥️Databases in PSTAT 10

In PSTAT 10 we will specifically work with **Relational Databases**.

- The most common databases used in industrial settings.
- Easy to interface with using **structured query language (SQL)**.
- Secure and very flexible design.
- Easy to maintain, been in use for over 50 years.



SQL Application Examples

Relational Database Model

🧩 Relational Database Model Parts

We divide the relational database model into three parts:

- Data structure
- Data integrity
- Data manipulation

> 📌 Today we discuss data structure and data integrity.

📖Core Definitions

We will consider the following definitions:

1. **Relation**: table of columns and rows (“table”)
2. **Attribute**: column of a relation
3. **Domain**: allowable attribute values
4. **Tuple**: row of a relation
5. **Keys** (Super, Candidate, Primary, Foreign): identifiers

Relational Databases

DEFINITION

A relational database is a collection of structured data organized into relations.

For example, see the table below:

Relations Example

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

🔎Interpretation

First Row

StudentId	FirstName	Course#
S1	Remy	PSTAT8

From the first row we see that a student with ID `S1`, named `Remy`, is enrolled in `PSTAT8`.

- What other information can we see?
- What other course is Remy enrolled in?
- Who does Remy share a course with?

🧩 Relations

Relations have the following properties:

- Order is irrelevant;
- Each row (tuple) must be unique;
- Every relation must have a unique name; and
- Relations can be updated (tuples added) as desired.

Relation Attributes

First Row

StudentId	FirstName	Course#
S1	Remy	PSTAT8

- The heading values are all attributes, i.e. `StudentId`, `FirstName`, `Course#`.
- The attribute values are in the columns.
- Rows are called tuples.
- Structurally relations are very similar to data frames in R.

🌐 Domains and Tuples

DEFINITION

The domain of a relation attribute is the set of atomic values used to model that attribute's data.

- Every relation attribute is linked to a domain.
- For example, the domain for `Course#` would be `PSTAT5A, PSTAT5LS, PSTAT8, PSTAT10, ...`

- Note that this is the set of all values a relation attribute can take — the values themselves do not have to be present in the relation.
- The tuples are all of the non-header rows.

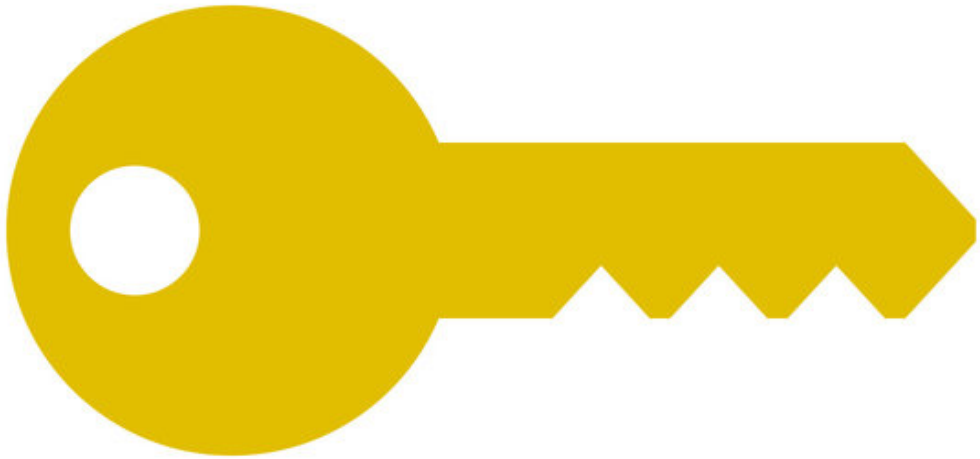
Keys

🔑 Keys

Keys are (subjectively) the **most important structural component** of the relational model.

DEFINITION

A **key** is a set of attributes that **determine the other attributes** in each tuple, **establish and identify links** between relations, and **set any required constraints** on a database.



🔑 Super Keys

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

DEFINITION

A **super key** is a set of attributes **uniquely identifying a tuple** in a relation.

💪 Exercise — Super Keys

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

Which of the following are super keys?

1. `(StudentId, FirstName, Course#)`
2. `(StudentId, FirstName)`
3. `(StudentId, Course#)`
4. `(StudentId)`

03:00

✅Solution — Super Keys

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

The following are both **super keys**:

1. `(StudentId, FirstName, Course#)`
2. `(StudentId, Course#)`

- A super key must **uniquely identify** every tuple.
- `Remy` (`S1`) appears in **two rows**, so `(StudentId)` and `(StudentId, FirstName)` are **not** super keys — they cannot tell those rows apart.
- Adding `Course#` **distinguishes** the tuples, so any set containing it uniquely identifies a row.

🔑 Candidate Keys

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

DEFINITION

A **candidate key** is a unique tuple identifier (super key) with **no redundancy**.

Consider the previous example's super keys:

`{StudentId, FirstName, Course#}` & `{StudentId, Course#}`

- These two keys encode (effectively) the **same information**.
- The `FirstName` attribute is **redundant**; both attribute sets point to the same tuple.
- Only `{StudentId, Course#}` is a candidate key.

🔑Primary Keys

DEFINITION

A primary key is a set of attributes that uniquely identifies any tuple in a given relation.

- Each relation has one primary key, selected from the candidate keys.
- The primary key is chosen to minimize the number of attributes.
- Primary keys cannot be duplicated, i.e. the same primary key cannot refer to multiple relations.
- Primary key values cannot be `null`.

> 📌In our previous example `{StudentId, Course#}` was the primary key for `ENROLLMENT`.

🔑 Foreign Keys

DEFINITION

A **foreign key** is a set of attributes that **links attributes across relations**.

- Specifically: given two relations `R1` and `R2`, the foreign key links attributes from `R1` to those in `R2`.
- This is **not intuitive** — we will look at an example.

Before we do though, think about why foreign keys might be **useful**?

- When would we want to link relations?
- What sort of structure does a relational DB take?

🔑 Example — Foreign Keys

`STUDENT` Relation

StudentId	FirstName
S1	Remy
S2	Sam
S3	Taylor
S4	Jordan

Primary Key: `{StudentId}`

`GPA` Relation

StudentId	GPA
S1	3.81
S2	3.97
S3	3.02
S4	2.79

Primary Key: `{StudentId, GPA}`

Foreign Key: `{StudentId}`

🔑 Key Summary

The structure of keys may remind you of the basics of set theory:

- Super Keys contain Candidate Keys.
- Candidate Keys contain Primary Keys.

Written in the language of set theory we have that:

$$\{\text{Super Keys}\} \supset \{\text{Candidate Keys}\} \supset \{\text{Primary Keys}\}$$

Foreign keys link relations together.

Data Integrity

🔒 Data Integrity

The first portion of this lecture was focused on data structure, i.e.:

- Organization of databases; and
- Technical vocabulary.

The next portion focuses on data integrity:

- The rules all relational databases must obey.
- Requirements to ensure consistency and completeness.
- Integrity rules:
    1. Entity
    2. Referential

🔒 Entity Integrity

DEFINITION

A relation has **entity integrity** when its primary keys are **unique** and have **values that are unique** and not `NULL` for every attribute (column).

> 📌 **Entity integrity** ensures we **do not have (potential) duplicate tuples** in a relation.

- How would duplicates impact **storage**?
- How would they impact **accessing data**?
- How would they impact the role of primary keys?

🔒 Example — Entity Integrity

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

Primary Key: `{StudentId, Course#}`

Suppose we want to add a new student named `Sam` who is enrolled in `PSTAT174`.

- Could we do this?
- If not, what would we still need?

Integrity Between Relations

We considered two types of keys:

- Super Keys, Candidate Keys, Primary Keys (all referring to a **single relation**)
- Foreign Keys (linking to **multiple relations**)

We think of **integrity** the same way:

- **Entity integrity** dealt with **individual relations**.
- **Referential integrity** constrains **relations between relations**.

 Referential Integrity

Foreign keys can take two types of values:

- They can be `NULL`.
- They can correspond to a specific primary key.

DEFINITION

Referential integrity ensures that we never attempt to reference a nonexistent tuple.

- Allows for easy cross-referencing across relations.
- Prevents calls to nonexistent data.

Example — Referential Integrity

`CUSTOMER` Relation

CUST_NO	NAME	Age
C1	Alex	21
C2	Charlie	26

Primary Key: `{CUST_NO}`

`SALES_ORDER` Relation

ORDER_NO	DATE	CUST_NO
011	6/11/23	C1
012	6/24/23	null
013	7/1/23	C3
015	7/12/23	C2

Foreign Key: `{CUST_NO}`

🔗 Example — Referential Integrity (Note)

Note the third tuple of the `SALES_ORDER` relation:

$$\{013,\ 7/1/23,\ C_3\}.$$

In customers, we have `CUST_NO` values:

$$\{C_1, C_2\}.$$

> 📌 Thus referential integrity is **violated** — we reference a primary key value which does not exist.

💪 Exercise — Evaluating Integrity

`ITEMS`

Product	Price ($)	Category	Manufacturer
Galaxy S23	899	Phones	Samsung
null	829	Phones	Apple
Macbook Air	1099	Laptops	Apple
Fire TV	449	Televisions	Amazon

`ORDERS`

Product	Order_NO	Cust_ID
Fire TV	1	CS1
Galaxy S23	2	CS11
Macbook Pro	3	CS11

We are given the tables above: `ITEMS` and `ORDERS`, with primary keys `Product` and `Cust_ID`.

1. What is the foreign key in `ORDERS`?
2. Is entity integrity violated? If so, how?
3. Is referential integrity violated? If so, how?

04:00

✅Solution — Evaluating Integrity

`ITEMS`

Product	Price ($)	Category	Manufacturer
Galaxy S23	899	Phones	Samsung
null	829	Phones	Apple
Macbook Air	1099	Laptops	Apple
Fire TV	449	Televisions	Amazon

`ORDERS`

Product	Order_NO	Cust_ID
Fire TV	1	CS1
Galaxy S23	2	CS11
Macbook Pro	3	CS11

1. `Product`
2. Yes; entity integrity fails — `null` primary key in `ITEMS` → `Product`.
3. Yes; referential integrity fails — `Macbook Pro` in `ORDERS` → `Product` has no matching tuple.

Schema

Relation Schema

DEFINITION

The relation schema is the relation name and its attributes.

StudentId	FirstName	Course#
S1	Remy	PSTAT8
S1	Remy	PSTAT131
S2	Sam	PSTAT10
S3	Taylor	PSTAT131
S4	Jordan	PSTAT174

Schema: `Enrollment {StudentId, FirstName, Course#}`

- Conventionally, the primary key would be indicated in some fashion.
- Recall that the primary key here was the pair `{StudentId, Course#}`; this type of key is known as a composite key.

📐 Database Schema

DEFINITION

The database schema is the set of all relation schema in a database.

- New relations can be added at any point.
- New relations must conform to integrity constraints.

For example, the earlier `STUDENT` and `GPA` relations together form a database schema:

```
Student {StudentId, FirstName},  GPA {StudentId, GPA}.
```

> 📌 The database schema records both the relations and the keys that link them.

Summary

✅Topics Covered

🙌This lecture we introduced ourselves to databases, specifically:

- The relational database model
- Keys
- Data Integrity; and
- Schema

17 Next Class

👉Next class we will discuss accessing and modifying databases.

- The database schema will form a core reference tool.
- We will often discuss database schema theoretically.